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Second-order integrators for Langevin equations with holonomic constraints

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Abstract

We propose a numerical scheme for the integration of the Langevin equation which is second-order accurate. More importantly, we indicate how to generalize this scheme to situations where holonomic constraints are added and show that the resulting scheme remains second-order accurate.
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1. Introduction

The evolution of a system of interacting particles in the presence of a thermal bath at temperature T can be described by the Langevin equation

$$M\ddot{x}(t) = F(x(t)) - \gamma M\dot{x}(t) + \sqrt{2k_B T\gamma}M^{1/2}\eta(t) \tag{1}$$

where $x(t) \in \mathbb{R}^{3N}$ denotes the position of the N particles, M the diagonal mass tensor, $F(x)$ the force, $\gamma > 0$ the friction coefficient and $\eta(t) = \dot{W}(t)$ is a white-noise ($W(t)$ being a Wiener process). As a result, the Langevin Eq. (1) is an useful tool to sample the Boltzmann–Gibbs distribution $\rho(x,v) = Z^{-1}\mathrm{e}^{-\beta H(x,v)}$ where $Z = \int_{\mathbb{R}^{3N}\times\mathbb{R}^{3N}} e^{-\beta H(x,v)}\mathrm{d}x\mathrm{d}v$ is the partition function and $H(x,v) = \frac{1}{2}\langle v, Mv\rangle + V(x)$ is the Hamiltonian (here we assumed that $F = -\nabla V$).

The question that we investigate here is how to generate an accurate approximation of a trajectory $(x(t), v(t))$ via time-discretization of (2). This question has been addressed by many authors. Among the most popular integrators are the one of van Gunsteren and Berendsen (vGB) proposed in [1], the one of Brooks–Brünger–Karplus (BBK) proposed in [2] and the Langevin impulse ($\widehat{\mathrm{LI}}$) integrator proposed in [3]. These various integrators have been recently reviewed and compared in [4], and it has been shown that BBK is first-order accurate, whereas both vGB and $\widehat{\mathrm{LI}}$ are second-order accurate. More recently, a class of integrators which are up to third-order accurate and also have some additional nice properties have been proposed in [5]. While all of these integrators have been derived for Langevin equations subject to no constraints, they have been often used to integrate systems in which holonomic constraints are present: this is usually done by simply applying an extra step of SHAKE [6] at every time-step. To the best of our knowledge, however, nobody has ever addressed the question of whether such a generalization of these integrators to simulate systems with constraints affect their numerical accuracy. This question is pertinent since higher-order accurate integrators allow for bigger time-steps and thereby permit to simulate the systems over longer time-intervals. In the deterministic context, because SHAKE does not affect the general structure of second-order integrators such as Verlet, one can easily show that it must preserve the order of accuracy of the integrator. The situation is more complex for Langevin integrators, however, because the structure of these integrators is more complicated and there is a nontrivial interplay between the force and the random noise at every time-step.

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In this Letter, we investigate the issue of the accuracy of a Langevin integrator used with SHAKE. Our main results are: (i) the integrator (21) and its quasi-symplectic version (23) in Section 2, which give simple and, to the best of our knowledge, new second-order accurate integrators for Langevin equations without constraints and (ii) the integrators (34) and (39) in Section 3, which show that SHAKE can be applied to (21) and (23) to impose holonomic constraints without loosing the second-order accuracy of the integrator. While we do not do so here, the method used in Section 3 can in principle be applied to prove (or disprove) that the popular integrators such as vGB, BBK or $\widehat{\text{LI}}$ keep their order of accuracy when applied in conjunction with SHAKE.

2. Basic algorithms

Written componentwise as a stochastic differential equation, (1) becomes

$$\begin{cases} \mathrm{d}x_i(t) = v_i(t)\mathrm{d}t \\ \mathrm{d}v_i(t) = (m_i^{-1}F_i(x(t)) - \gamma_i v_i(t))\mathrm{d}t + \sqrt{2k_{\mathrm{B}}T\gamma_i m_i^{-1}}\mathrm{d}W_i(t) \end{cases} \tag{2}$$

where (x_1, x_2, x_3) are the x, y and z components of particle 1 (v_1, v_2, v_3) are the components of its velocity, $m_1 = m_2 = m_3 > 0$ is its mass and $\gamma_1 = \gamma_2 = \gamma_3 > 0$ is the friction coefficient it is subject to, and similarly for (x_4, x_5, x_6), etc. To simplify the formula, it will be convenient to write (2) as

$$\begin{cases} \mathrm{d}x_i(t) = v_i(t)\mathrm{d}t \\ \mathrm{d}v_i(t) = (f_i(x(t)) - \gamma_i v_i(t))\mathrm{d}t + \sigma_i \mathrm{d}W_i(t), \end{cases} \tag{3}$$

where we defined $f_i(x) = m_i^{-1}F_i(x)$ and $\sigma_i = \sqrt{2k_{\mathrm{B}}T\gamma_i m_i^{-1}}$. Going back to vectorial notations, this equation can be written as the system of integral equations

$$\begin{cases} x(t+h) = x(t) + \int_t^{t+h} v(s)\mathrm{d}s \\ v(t+h) = v(t) + \int_t^{t+h} f(x(s))\mathrm{d}s - \gamma\int_t^{t+h} v(s)\mathrm{d}s \\ \qquad\quad +\sigma(W(t+h) - W(h)) \end{cases} \tag{4}$$

Using $x(s) = x(t) + O(h)$ and $v(s) = v(t) + O(\sqrt{h})$ (recall that $W(t+h) - W(h) = O(\sqrt{h})$) under the integrals and neglecting the corrections of order $h^{3/2}$ and higher, we arrive at the Euler–Maruyama scheme [7,8]:

$$\begin{cases} x^{n+1} = x^n + hv^n \\ v^{n+1} = v^n + hf(x^n) - h\gamma v^n + \sqrt{h}\sigma\xi^n. \end{cases} \tag{5}$$

where x^n is the numerical approximation of $x(nh)$, v^n that of $v(nh)$ and ξ^n are independent (for different n) Gaussian variables with mean zero and covariance $\mathbb{E}(\xi_i^n\xi_j^n) = \delta_{ij}$ (here we used $W(t+h) - W(t) \stackrel{d}{=} \sqrt{h}\xi^n$, where $\stackrel{d}{=}$ denotes equality in law or distribution). It can be shown [7,8] that the scheme (5) is first-order accurate, meaning that, given any suitable observable ϕ,

$$\sup_{0\leqslant n\leqslant T/h} |\mathbb{E}\phi(x(nh), v(nh)) - \mathbb{E}\phi(x^n, v^n)| \leqslant Ch \tag{6}$$

where $(x(nh), v(nh))$ is the (exact) solution of (2) evaluated at $t = nh$, (x^n, v^n) is its numerical approximation by (5) and it is assumed that $(x(0), v(0)) = (x^0, v^0)$; $\mathbb{E}$ denotes expectation with respect to the noises in (2) and (5), and the constant C depends on $T < \infty$ and the observable $\phi(x, v)$ but not on h. Note that (6) is a finite-time error estimate, hence it says little about the error in computing equilibrium averages, but it indicates that it may be necessary to use (5) with a very small time-step h when the force field is stiff. Hence (5) may be insufficient in applications (Fig. 1).

To do better than (5), go back to (4), and for $v(s)$ in the integral $\int_t^{t+h} v(s)\mathrm{d}s$ use

$$v(s) = v(t) + \int_t^s f(x(u))\mathrm{d}u - \gamma\int_t^s v(u)\mathrm{d}u + \sigma(W(s) - W(t)) \tag{7}$$

Using $x(u) = x(t) + O(h)$ and $v(u) = v(t) + O(\sqrt{h})$ at this level we deduce that (recall that $s \in [t, t+h]$ and hence $s - t = O(h)$)

$$\begin{aligned} v(s) = v(t) &+ (s-t)(f(x(t)) - \gamma v(t)) \\ &+ \sigma(W(s) - W(t)) + O(h^{3/2}) \end{aligned} \tag{8}$$

and hence

$$\begin{aligned} \int_t^{t+h} v(s)\mathrm{d}s = hv(t) &+ \tfrac{1}{2}h^2(f(x(t)) - \gamma v(t)) \\ &+ \sigma\int_t^{t+h}(W(s) - W(t))\mathrm{d}s + O(h^{5/2}) \end{aligned} \tag{9}$$

Similarly, for $f(x(s))$ in the integral $\int_t^{t+h} f(x(s))\mathrm{d}s$, use

$$f(x(s)) = f(x(t)) + \int_t^s v(u)\cdot\nabla f(x(u))\mathrm{d}u \tag{10}$$

then $x(u) = x(t) + O(h)$ and $v(u) = v(t) + O(\sqrt{h})$ to arrive at

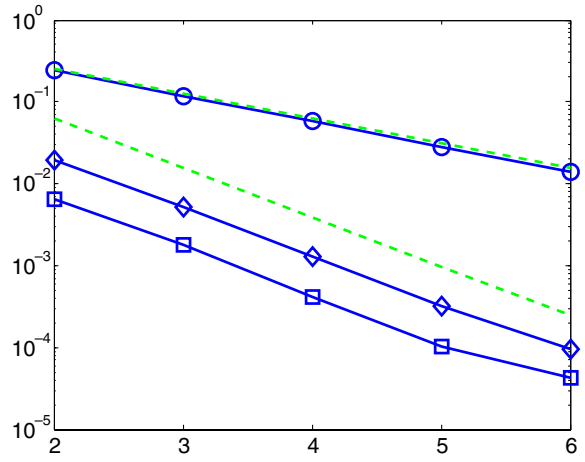


Fig. 1. Numerical errors of the integrators (5), (21) and (23) on the example of $\dot{x}(t) = v(t)$, $\dot{v}(t) = -x(t) - v(t) + \sqrt{2}\eta(t)$, $x(0) = v(0) = 0$. The times steps are $h = 2^{-n}$ for $n = 2,3,4,5,6$. The quantity monitored for the error is the estimate of $\mathbb{E}(x^2(1) + v^2(1)) = 0.9796111900\ldots$ computed by ensemble averaging over 10^8 independent realizations. The dashed curves are the graphs of the functions $2^{-n}(=h)$ and $2^{-2n}(=h^2)$ versus n. The circles are the errors obtained using the first-order integrator (5) and they are consistent with the error estimate (6); the diamonds are the errors obtained using the second-order integrator (21) and they are consistent with the error estimate (18); the squares are the errors obtained using the quasi-symplectic second-order integrator (23).

$$f(x(s)) = f(x(t)) + (t-s)v(t)\cdot\nabla f(x(t)) + O(h^{3/2}) \tag{11}$$

and

$$\int_t^{t+h} f(x(s))\mathrm{d}s = hf(x(t)) + \tfrac{1}{2}h^2 v(t)\cdot\nabla f(x(t)) + O(h^{5/2}) \tag{12}$$

Recalling that $\mathbb{E}(W_i(s)W_j(s')) = \delta_{ij}\min(s,s')$, one sees that for fixed t and h, $\int_t^{t+h}(W(s)-W(t))\mathrm{d}s$ is a Gaussian variable with mean zero and such that

$$\begin{aligned}&\mathbb{E}\left(\int_t^{t+h}(W_i(s)-W_i(t))\mathrm{d}s\right)\left(\int_t^{t+h}(W_j(s)-W_j(t))\mathrm{d}s\right) = \tfrac{1}{3}h^3\delta_{ij}\\ &\mathbb{E}((W_i(t+h)-W_i(t))\int_t^{t+h}(W_j(s)-W_j(t))\mathrm{d}s) = \tfrac{1}{2}h^2\delta_{ij}\end{aligned} \tag{13}$$

Since $(W(t+h)-W(t))$ is also a Gaussian variable with mean zero and covariance $h\delta_{ij}$, this means that

$$\begin{aligned}&W(t+h)-W(t) \overset{d}{=} \sqrt{h}\xi^n\\ &\int_t^{t+h}(W(s)-W(t))\mathrm{d}s \overset{d}{=} h^{3/2}\left(\tfrac{1}{2}\xi^n + \tfrac{1}{2\sqrt{3}}\eta^n\right)\end{aligned} \tag{14}$$

where (ξ^n,η^n) are independent Gaussian variables with mean zero and covariance

$$\mathbb{E}(\xi_i^n\xi_j^n) = \mathbb{E}(\eta_i^n\eta_j^n) = \delta_{ij}, \quad \mathbb{E}(\xi_i^n\eta_j^n) = 0 \tag{15}$$

Inserting (9) and (12) into (4), neglecting terms of order $h^{5/2}$ and higher and using (14) we arrive at the integrator

$$\begin{cases} x^{n+1} = x^n + hv^n + A^n\\ v^{n+1} = v^n + hf(x^n) - h\gamma v^n + \sqrt{h}\sigma\xi^n + \tfrac{1}{2}h^2 v^n\cdot\nabla f(x^n) - \gamma A^n\end{cases} \tag{16}$$

where

$$A^n = \tfrac{1}{2}h^2(f(x^n)-\gamma v^n) + \sigma h^{3/2}\left(\tfrac{1}{2}\xi^n + \tfrac{1}{2\sqrt{3}}\eta^n\right) \tag{17}$$

This integrator is in fact a special case of Milstein's scheme [7,8] and it is second-order accurate, i.e. we now have (compare (6))

$$\sup_{0\leqslant n\leqslant T/h} |\mathbb{E}\phi(x(nh),v(nh)) - \mathbb{E}\phi(x^n,v^n)| \leqslant Ch^2 \tag{18}$$

where (x^n,v^n) is the solution of (16). However, (16) is unpleasant because it involves the term $v^n\cdot\nabla f(x^n)$. To get rid of the latter, notice that (16) implies that

$$f(x^{n+1}) = f(x^n) + hv^n\cdot\nabla f(x^n) + O(h^{3/2}) \tag{19}$$

or, equivalently, that

$$\tfrac{1}{2}h^2 v^n\cdot\nabla f(x^n) = \tfrac{1}{2}h(f(x^{n+1})-f(x^n)) + O(h^{5/2}). \tag{20}$$

Using (20) in (16) we can rewrite this integrator as

$$\begin{cases} x^{n+1} = x^n + hv^n + A^n\\ v^{n+1} = v^n + \tfrac{1}{2}h(f(x^{n+1})+f(x^n)) - h\gamma v^n + \sqrt{h}\sigma\xi^n - \gamma A^n\end{cases} \tag{21}$$

To the best of our knowledge, the integrator (21) is new: it is explicit and, like (16), it is second-order accurate, i.e. (18) still holds now with (x^n,v^n) being the solution of (21). It is a generalization to the Langevin equation of the familiar velocity-Verlet algorithm, which is obtained from (21) by setting $\gamma=\sigma=0$ (in which case (2) reduces to Hamilton's equation of motion)

$$\begin{cases} x^{n+1} = x^n + hv^n + \tfrac{1}{2}h^2 f(x^n)\\ v^{n+1} = v^n + \tfrac{1}{2}h(f(x^{n+1})+f(x^n))\end{cases} \tag{22}$$

Notice that (21) requires no more force field evaluation than (22), and no more than (5).

There is another way to modify the integrator (21) without losing either accuracy or simplicity (i.e. such that the computation of $\nabla f(x^n)$ is avoided and each time step requires only one force field evaluation):

$$\begin{cases} v^{n+1/2} = v^n + \tfrac{1}{2}hf(x^n) - \tfrac{1}{2}h\gamma v^n + \tfrac{1}{2}\sqrt{h}\sigma\xi^n\\ \qquad\qquad - \tfrac{1}{8}h^2\gamma(f(x^n)-\gamma v^n) - \tfrac{1}{4}h^{3/2}\gamma\sigma\left(\tfrac{1}{2}\xi^n + \tfrac{1}{\sqrt{3}}\eta^n\right)\\ x^{n+1} = x^n + hv^{n+1/2} + h^{3/2}\sigma\tfrac{1}{2\sqrt{3}}\eta^n\\ v^{n+1} = v^{n+1/2} + \tfrac{1}{2}hf(x^{n+1}) - \tfrac{1}{2}h\gamma v^{n+1/2} + \tfrac{1}{2}\sqrt{h}\sigma\xi^n\\ \qquad\qquad - \tfrac{1}{8}h^2\gamma(f(x^{n+1})-\gamma v^{n+1/2}) - \tfrac{1}{4}h^{3/2}\gamma\sigma\left(\tfrac{1}{2}\xi^n + \tfrac{1}{\sqrt{3}}\eta^n\right)\end{cases} \tag{23}$$

It can be checked by direct comparison that x^{n+1} and v^{n+1} from (23) agree up to order h^2 with x^{n+1} and v^{n+1} from either (16) or (21). To the best of our knowledge, the integrator (23) is also new: it is a second-order generalization of the BBK integrator [2]. Setting $\gamma=\sigma=0$ in (23), this integrator also reduces to velocity-Verlet, now written as

$$\begin{cases} v^{n+1/2} = v^n + \tfrac{1}{2}hf(x^n)\\ x^{n+1} = x^n + hv^{n+1/2}\\ v^{n+1} = v^{n+1/2} + \tfrac{1}{2}hf(x^{n+1})\end{cases} \tag{24}$$

The integrator (23) may have some advantage over (21) because, unlike (21), (23) is quasi-symplectic in the sense of [5]: it reduces to a symplectic integrator when $\gamma=\sigma=0$ and the Jacobian J of the transformation from (x^n,v^n) to (x^{n+1},v^{n+1}) is independent of (x^n,v^n). A direct calculation gives $J = \prod_{i=1}^{3N}\left(1-\tfrac{1}{2}h\gamma_i + \tfrac{1}{8}h^2\gamma_i^2\right)^2$, to be compared with the exact value $\exp(-h\sum_{i=1}^{3N}\gamma_i)$.

3. Generalizations to situations with constraints

Many situations require to generalize the Langevin Eq. (2) to include hard constraints. Let us assume that Λ constraints are given in the form of $\theta_1(x)=\cdots=\theta_\Lambda(x)=0$. Then (2) becomes

$$\begin{cases}\dot{x}(t) = v(t)\\ \dot{v}(t) = f(x(t)) - \gamma v(t) + \sigma\eta(t) - \sum_{\alpha=1}^{\Lambda}\lambda_\alpha(t)\nabla\theta_\alpha(x(t))\end{cases} \tag{25}$$

where $\lambda_1(t),\ldots,\lambda_\Lambda(t)$ are Lagrange multiplier added to satisfy the constraints. Since $\theta_\alpha(x(t))=0$ for all $\alpha=1,\ldots,\Lambda$ and all t, it follows that

$$0 = \dot{\theta}_\alpha(x(t)) = v(t) \cdot \nabla\theta_\alpha(x(t)) \tag{26}$$

which indicates that at all times $v(t)$ must be orthogonal to all the $\nabla\theta_\alpha(x(t))$ (and so the initial velocity $v(0)$ must also satisfy this condition). Similarly, we must have

$$\begin{aligned} 0 = \ddot{\theta}_\alpha(x(t)) &= v(t)v(t) : \nabla\nabla\theta_\alpha(x(t)) + \dot{v}(t) \cdot \nabla\theta_\alpha(x(t)) \\ &= v(t)v(t) : \nabla\nabla\theta_\alpha(x(t)) + (f(x(t)) - \gamma v(t) + \sigma\eta(t)) \cdot \nabla\theta_\alpha(x(t)) \\ &\quad - \sum_{\beta=1}^{\Lambda} \lambda_\beta(t) \nabla\theta_\beta(x(t)) \cdot \nabla\theta_\alpha(x(t)), \end{aligned} \tag{27}$$

where, given $a, b \in \mathbb{R}^{3N}$, $ab : \nabla\nabla = \sum_{i,j=1}^{3N} a_i b_j \partial^2/\partial x_i \partial x_j$ (27) is an equation for $\lambda_\alpha(t)$. Letting

$$K_{\alpha\beta}(x) = \nabla\theta_\alpha(x) \cdot \nabla\theta_\beta(x) \in \mathbb{R}^\Lambda \times \mathbb{R}^\Lambda \tag{28}$$

and assuming that this matrix is invertible (27) can be solved in $\lambda_\alpha(t)$ as

$$\begin{aligned} \lambda_\alpha(t) = &\sum_{\beta=1}^{\Lambda} K^{-1}_{\alpha\beta}(x(t)) \nabla\theta_\beta(x(t)) \cdot (f(x(t)) - \gamma v(t) + \sigma\eta(t)) \\ &+ \sum_{\beta=1}^{\Lambda} K^{-1}_{\alpha\beta}(x(t)) \nabla\nabla\theta_\beta(x(t)) : v(t)v(t) \end{aligned} \tag{29}$$

Inserting (29) into (25), one sees that this equation can be written explicitly as

$$\begin{cases} \dot{x}(t) = v(t) \\ \dot{v}(t) = P(x(t))(f(x(t)) - \gamma v(t) + \sigma\eta(t)) \\ \qquad - \sum_{\alpha,\beta=1}^{\Lambda} \nabla\theta_\alpha(x(t)) K^{-1}_{\alpha\beta}(x(t)) \nabla\nabla\theta_\beta(x(t)) : v(t)v(t) \end{cases} \tag{30}$$

where $P(x)$ is the orthogonal projector unto the manifold $\theta_1(x) = \cdots = \theta_\Lambda(x) = 0$: componentwise, it is given by

$$P_{ij}(x) = \delta_{ij} - \sum_{\alpha,\beta=1}^{\Lambda} K^{-1}_{\alpha\beta}(x) \frac{\partial\theta_\alpha(x)}{\partial x_i} \frac{\partial\theta_\beta(x)}{\partial x_j} \tag{31}$$

Since the noise $\eta(t)$ is multiplied by a function of $x(t)$ alone in (30), there is no problem of interpretation with this equation and the heuristic way we arrived at (30) can be justified rigorously. Notice that $\lambda_\alpha(t)$ contains a martingale part,

$$\mathrm{d}\lambda^m_\alpha(t) = \sigma \sum_{\beta=1}^{\Lambda} K^{-1}_{\alpha\beta}(x(t)) \nabla\theta_\beta(x(t)) \cdot \mathrm{d}W(t). \tag{32}$$

and a bounded variation part

$$\begin{aligned} \mathrm{d}\lambda^b_\alpha(t) = &\sum_{\beta=1}^{\Lambda} K^{-1}_{\alpha\beta}(x(t)) \nabla\theta_\beta(x(t)) \cdot (f(x(t)) - \gamma v(t))\mathrm{d}t \\ &+ \sum_{\beta=1}^{\Lambda} K^{-1}_{\alpha\beta}(x(t)) \nabla\nabla\theta_\beta(x(t)) : v(t)v(t)\mathrm{d}t \end{aligned} \tag{33}$$

We could write down directly a numerical integrator for (30) by proceeding as in Section 2. This, however, would have two disadvantages. First, the resulting integrator would be rather complicated and, e.g. involve second-order derivatives of $\theta_\alpha(x)$, which are cumbersome to compute. Second, and more importantly, even if we write down a second-order accurate integrator, this integrator would only conserve the constraints $\theta_1(x(t)) = \ldots = \theta_\Lambda(x(t)) = 0$ to $O(h^2)$. This is very unsatisfactory since over long times, the constraints would then drift away.

In the spirit of SHAKE [6], it is much more convenient to write down an integrator which is accurate to second-order but conserves the constraints exactly. Below, we show that the following integrator satisfies these conditions:

$$\begin{cases} x^{n+1} = x^n + hv^n + A^n - \sum_{\alpha=1}^{\Lambda} \mu^n_\alpha \nabla\theta_\alpha(x^n) \\ v^{n+1} = v^n + \frac{1}{2}h(f(x^{n+1}) + f(x^n)) - h\gamma v^n + \sqrt{h}\sigma\xi^n - \gamma A^n \\ \qquad -h^{-1} \sum_{\alpha=1}^{\Lambda} (\mu^*_\alpha \nabla\theta_\alpha(x^{n+1}) + \mu^n_\alpha \nabla\theta_\alpha(x^n)) \end{cases} \tag{34}$$

Here A^n is given in (17) $(\mu^n_1, \ldots, \mu^n_\Lambda)$ are chosen so that

$$\theta_\alpha(x^{n+1}) = 0, \quad \alpha = 1, \ldots, \Lambda \tag{35}$$

and $(\mu^*_1, \ldots, \mu^*_\Lambda)$ are chosen so that

$$v^{n+1} \cdot \nabla\theta_\alpha(x^{n+1}) = 0, \quad \alpha = 1, \ldots, \Lambda \tag{36}$$

(34) simply means that we can apply RATTLE to (21), remain second-order accurate and satisfy the constraints (26) exactly. The multipliers μ^n_α can be obtained via SHAKE; the multipliers μ^*_α can be obtained via solution of a linear system. For future reference, notice that the updating step for the velocities in (34) can be written as

$$v^{n+1} = P(x^{n+1})v^* \tag{37}$$

where

$$\begin{aligned} v^* = &v^n + \tfrac{1}{2}h(f(x^{n+1}) + f(x^n)) - h\gamma v^n + \sqrt{h}\sigma\xi^n - \gamma A^n \\ &- h^{-1} \sum_{\alpha=1}^{\Lambda} \mu^n_\alpha \nabla\theta_\alpha(x^n). \end{aligned} \tag{38}$$

In the same way, we can apply RATTLE to (23), remain second-order accurate and satisfy the constraints (26) exactly. In this case, the integrator reads:

$$\begin{cases} v^{n+1/2} = v^n + \frac{1}{2}hf(x^n) - \frac{1}{2}h\gamma v^n + \frac{1}{2}\sqrt{h}\sigma\xi^n \\ \qquad - \frac{1}{8}h^2\gamma(f(x^n) - \gamma v^n) - \frac{1}{4}h^{3/2}\gamma\sigma\left(\frac{1}{2}\eta^n + \frac{1}{\sqrt{3}}\eta^n\right) \\ \qquad -h^{-1} \sum_{\alpha=1}^{\Lambda} \mu^n_\alpha \nabla\theta_\alpha(x^n) \\ x^{n+1} = x^n + hv^{n+1/2} + h^{3/2}\sigma\frac{1}{2\sqrt{3}}\eta^n \\ v^{n+1} = v^{n+1/2} + \frac{1}{2}hf(x^{n+1}) - \frac{1}{2}h\gamma v^{n+1/2} + \frac{1}{2}\sqrt{h}\sigma\xi^n \\ \qquad - \frac{1}{8}h^2\gamma(f(x^{n+1}) - \gamma v^{n+1/2}) - \frac{1}{4}h^{3/2}\gamma\sigma\left(\frac{1}{2}\eta^n + \frac{1}{\sqrt{3}}\eta^n\right) \\ \qquad -h^{-1} \sum_{\alpha=1}^{\Lambda} \mu^*_\alpha \nabla\theta_\alpha(x^{n+1}) \end{cases} \tag{39}$$

where the Lagrange multipliers $(\mu^n_1, \ldots, \mu^n_\Lambda)$ and $(\mu^*_1, \ldots, \mu^*_\Lambda)$ are chosen so that (35) and (37) are satisfied. It can be checked by direct comparison that x^{n+1} and v^{n+1} from (39) agree up to order h^2 with x^{n+1} and v^{n+1} from (34).

The remainder of this Letter is devoted to show the consistency of (34) with respect to (30) and establish that it is second-order accurate; this also establishes the consistency and second-order accuracy of (39) since this integrator coincide with (34) to $O(h^2)$. Since we were unable to find a soft argument establishing this claim, we will proceed the hard way, write down an integrator for (30) which is second-order accurate and show that, to $O(h^2)$ this integrator is equivalent to (34). For simplicity, we will treat the simpler case where there is a single constraint, $\Lambda = 1$ and $\theta_1(x) = \theta(x)$ (the general case can be treated similarly but the algebra become even more tedious). Under this assumption, (30) can be written as the following integral equation (using $P(x(t))v(t) = v(t)$ which follows from (26))

$$\begin{cases} x(t+h) = x(t) + \int_t^{t+h} v(s)\mathrm{d}s \\ v(t+h) = v(t) + \int_t^{t+h}(P(x(s))f(x(s)) - \gamma v(s))\mathrm{d}s \\ \qquad +\sigma\int_t^{t+h} P(x(s))\mathrm{d}W(s) \\ \qquad -\int_t^{t+h} \nabla\theta(x(t))|\nabla\theta(x(s))|^{-2}\nabla\nabla\theta(x(s)) \\ \qquad : v(s)v(s)\mathrm{d}s. \end{cases} \tag{40}$$

Proceeding as in Section 2, we can integrate once and replace $v(s)$, etc. by their value according to (40), then expand and keep only the term of order h^2 or lower. The relevant expressions are (here for simplicity we denote $x(t) \equiv x$ and $v(t) \equiv v$)

$$\begin{aligned} \int_t^{t+h} v(s)\mathrm{d}s &= hv + \tfrac{1}{2}h^2(P(x)f(x) - \gamma v) \\ &\quad + \sigma P(x)\int_t^{t+h}(W(s) - W(t))\mathrm{d}s \\ &\quad - \tfrac{1}{2}h^2|\nabla\theta(x)|^{-2}(\nabla\nabla\theta(x) : vv)\nabla\theta(x) \\ &\quad + O(h^{5/2}) \end{aligned} \tag{41}$$

$$\begin{aligned} \int_t^{t+h} P(x(s))f(x(s))\mathrm{d}s &= hP(x)f(x) \\ &\quad + \tfrac{1}{2}h^2 v\cdot\nabla(P(x)f(x)) + O(h^{5/2}) \end{aligned} \tag{42}$$

$$\begin{aligned} &\int_t^{t+h} \nabla\theta(x(s))|\nabla\theta(x(s))|^{-2}\nabla\nabla\theta(x(s)) : v(s)v(s)\mathrm{d}s \\ &\quad = h|\nabla\theta(x)|^{-2}(\nabla\nabla\theta(x) : vv)\nabla\theta(x) \\ &\qquad + \tfrac{1}{2}h^2 v\cdot\nabla(\nabla\theta(x)|\nabla\theta(x)|^{-2}\nabla\nabla\theta(x)) : vv \\ &\qquad + h^2|\nabla\theta(x)|^{-2}(\nabla\nabla\theta(x) : v(P(x)f(x) - \gamma v)\nabla\theta(x)) \\ &\qquad + 2\sigma|\nabla\theta(x)|^{-2}(\nabla\nabla\theta(x) : vP(x)\int_t^{t+h}(W(s) - W(t))\mathrm{d}s)\nabla\theta(x) \\ &\qquad - h^2|\nabla\theta(x)|^{-4}(\nabla\nabla\theta(x) : v\nabla\theta(x))(\nabla\nabla\theta(x) : vv)\nabla\theta(x) \\ &\qquad + \tfrac{1}{2}h^2\sigma^2|\nabla\theta(x)|^{-2}(P(x) : \nabla\nabla\theta(x))\nabla\theta(x) + O(h^{5/2}) \end{aligned} \tag{43}$$

where we used (this identity holds almost surely under iterations within the scheme)

$$\begin{aligned} &\nabla\nabla\theta(x(s)) : (P(x(s))\int_t^s \mathrm{d}W(u))(P(x(s))\int_t^s \mathrm{d}W(u)) \\ &\qquad = (s-t)P(x(s)) : \nabla\nabla\theta(x(s)) \end{aligned} \tag{44}$$

and, finally,

$$\begin{aligned} \int_t^{t+h} P(x(s))\mathrm{d}W(s) &= P(x)(W(t+h) - W(t)) \\ &\quad + v\cdot\nabla P(x)\int_t^{t+h}(s-t)\mathrm{d}W(s) + O(h^{5/2}) \end{aligned} \tag{45}$$

Using $\int_t^{t+h}(s-t)\mathrm{d}W(s) = h(W(t+h) - W(t)) - \int_t^{t+h}(W(s) - W(t))\mathrm{d}s$ together with (14), inserting these expressions in (40) and neglecting terms of order $h^{5/2}$ and higher gives the following integrator

$$\begin{cases} x^{n+1} = x^n + hv^n + P^nA^n - \tfrac{1}{2}h^2|\nabla\theta^n|^{-2}(v^nv^n : \nabla\nabla\theta^n)\nabla\theta^n \\ v^{n+1} = v^n + hP^nf^n - h\gamma v^n + \sqrt{h}\sigma P^n\xi^n \\ \qquad -h|\nabla\theta^n|^{-2}(v^nv^n : \nabla\nabla\theta^n)\nabla\theta^n + \tfrac{1}{2}h^2 v^n\cdot\nabla(P^nf^n) \\ \qquad -\gamma P^nA^n + \tfrac{1}{2}h^2\gamma|\nabla\theta^n|^{-2}(v^nv^n : \nabla\nabla\theta^n)\nabla\theta^n \\ \qquad +h^{3/2}\sigma(v^n\cdot\nabla P^n)\left(\tfrac{1}{2}\xi^n - \tfrac{1}{2\sqrt{3}}\eta^n\right) \\ \qquad -\tfrac{1}{2}h^2 v^n\cdot\nabla(|\nabla\theta^n|^{-2}(v^nv^n : \nabla\nabla\theta^n)\nabla\theta^n) \\ \qquad -2|\nabla\theta^n|^{-2}(v^n(P^nA^n) : \nabla\nabla\theta^n)\nabla\theta^n \\ \qquad +h^2|\nabla\theta^n|^{-4}(v^nv^n : \nabla\nabla\theta^n)(v^n\nabla\theta^n : \nabla\nabla\theta^n)\nabla\theta^n \\ \qquad -\tfrac{1}{2}h^2\sigma^2|\nabla\theta^n|^{-2}(P^n : \nabla\nabla\theta^n)\nabla\theta^n \end{cases} \tag{46}$$

where $f^n = f(x^n)$, $\theta^n = \theta(x^n)$, $P^n = P(x^n)$. Notice that (46) is second-order accurate, but unlike (34), this integrator does not preserve the constraints exactly: in fact $\theta^n = O(h^2)$ and $P^nv^n = v^n + O(h^2)$.

Let us now show that (34) is identical to (46) to $O(h^2)$. To this end, notice first that, using (34) and (35) implies that (using $\theta^n = 0$ and $v^n\cdot\nabla\theta^n = 0$)

$$\begin{aligned} 0 = \theta(x^{n+1}) &= \theta(x^n + hv^n + A^n - \mu^n\nabla\theta^n) \\ &= A^n\cdot\nabla\theta^n - \mu^n|\nabla\theta^n|^2 \\ &\quad + \tfrac{1}{2}(hv^n - \mu^n\nabla\theta^n)(hv^n - \mu^n\nabla\theta^n) : \nabla\nabla\theta^n + O(h^{5/2}) \end{aligned} \tag{47}$$

As a result

$$\begin{aligned} \mu^n &= |\nabla\theta^n|^{-2}A^n\cdot\nabla\theta^n + \tfrac{1}{2}|\nabla\theta^n|^{-2}(hv^n - \mu^n\nabla\theta^n) \\ &\quad \times (hv^n - \mu^n\nabla\theta^n) : \nabla\nabla\theta^n + O(h^{5/2}) \\ &= |\nabla\theta^n|^{-2}A^n\cdot\nabla\theta^n + \tfrac{1}{2}h^2|\nabla\theta^n|^{-2}v^nv^n : \nabla\nabla\theta^n + O(h^{5/2}) \end{aligned} \tag{48}$$

(Notice for later use that even though μ^n is divided by h in the equation for v^{n+1} in (34), we do not need to compute terms up to $O(h^3)$ because v^{n+1} is obtained via (37) and $P^{n+1}\nabla\theta^n = O(h)$.)

(48) implies that the way the positions are updated in (34) is explicitly

$$\begin{aligned} x^{n+1} &= x^n + hv^n + A^n - |\nabla\theta^n|^{-2}A^n\cdot\nabla\theta^n\nabla\theta^n - \tfrac{1}{2}h^2|\nabla\theta^n|^{-2} \\ &\quad \times (v^nv^n : \nabla\nabla\theta^n)\nabla\theta^n + O(h^{5/2}) \\ &= x^n + hv^n + P^nA^n - \tfrac{1}{2}h^2|\nabla\theta^n|^{-2}(v^nv^n : \nabla\nabla\theta^n)\nabla\theta^n + O(h^{5/2}) \end{aligned} \tag{49}$$

which is consistent to second-order with the equation for x^{n+1} in (46).

Consider next the way the velocities are updated in (34). In the present case where $\Lambda = 1$ (38) is explicitly (using (48))

$$\begin{aligned} v^* &= v^n + \tfrac{1}{2}h(f^{n+1} + f^n) - h\gamma v^n + \sqrt{h}\sigma\xi^n - \gamma A^n - h^{-1}\mu^n\nabla\theta^n \\ &= v^n + hf^n + \tfrac{1}{2}h^2 v^n\cdot\nabla f^n - h\gamma v^n + \sqrt{h}\sigma\xi^n - \gamma A^n \\ &\quad - h^{-1}|\nabla\theta^n|^{-2}(\nabla\theta^n\cdot A^n)\nabla\theta^n - \tfrac{1}{2}h|\nabla\theta^n|^{-2} \\ &\quad \times (v^nv^n : \nabla\nabla\theta^n)\nabla\theta^n + O(h^{3/2}) \end{aligned} \tag{50}$$

where the terms of order $h^{3/2}$ and h^2 come from $-h^{-1}\mu^n\nabla\theta^n$ and hence do not contribute to v^{n+1} since $P^{n+1}\nabla\theta^n = O(h)$

(see the remark after (48)). On the other hand, using (49), we also have that

$$\begin{aligned} P^{n+1}v^* = {} & P^n v^* + h v^n \cdot \nabla P^n v^* + P^n A^n \cdot \nabla P^n v^* \\ & - \tfrac{1}{2} h^2 |\nabla\theta^n|^{-2} (v^n v^n : \nabla\nabla\theta^n) \nabla\theta^n \cdot \nabla P^n v^* \\ & + \tfrac{1}{2} h^2 v^n v^n : \nabla\nabla P^n v^* + O(h^{5/2}) \end{aligned} \tag{51}$$

Combining (48), (50) and (51), we deduce that

$$\begin{aligned} v^{n+1} = {} & v^n + h P^n f^n + \tfrac{1}{2} h^2 P^n v^n \cdot \nabla f^n - h\gamma v^n + \sqrt{h}\sigma P^n \xi^n \\ & - \gamma P^n A^n + h v^n \cdot \nabla P^n v^n + h^2 v^n \cdot \nabla P^n f^n - h^2 \gamma v^n \cdot \nabla P^n v^n \\ & + h^{3/2} \sigma v^n \cdot \nabla P^n \xi^n - |\nabla\theta^n|^{-2} (\nabla\theta^n \cdot A^n) v^n \cdot \nabla P^n \nabla\theta^n \\ & - \tfrac{1}{2} h^2 |\nabla\theta^n|^{-2} (v^n v^n : \nabla\nabla\theta^n) v^n \cdot \nabla P^n \nabla\theta^n + P^n A^n \cdot \nabla P^n v^n \\ & + h^2 \sigma^2 P^n \left(\tfrac{1}{2}\xi^n + \tfrac{1}{2\sqrt{3}} \eta^n \right) \cdot \nabla P^n \xi^n - h^2 \sigma^2 |\nabla\theta^n|^{-2} \\ & \times \left(\left(\tfrac{1}{2}\xi^n + \tfrac{1}{2\sqrt{3}} \eta^n \right) \cdot \nabla\theta^n \right) P^n \left(\tfrac{1}{2}\xi^n + \tfrac{1}{2\sqrt{3}} \eta^n \right) \cdot \nabla P^n \nabla\theta^n \\ & - \tfrac{1}{2} h^2 |\nabla\theta^n|^{-2} (v^n v^n : \nabla\nabla\theta^n) \nabla\theta^n \cdot \nabla P^n v^n \\ & + \tfrac{1}{2} h^2 v^n v^n : \nabla\nabla P^n v^n + O(h^{5/2}) \end{aligned} \tag{52}$$

To simplify this expression, let us derive first a few useful identities. Given any two vectors $a, b \in \mathbb{R}^{3N}$, we have

$$\begin{aligned} a \cdot \nabla P^n b = {} & - |\nabla\theta^n|^{-2} (ab : \nabla\nabla\theta^n) \nabla\theta^n \\ & - |\nabla\theta^n|^{-2} (b \cdot \nabla\theta^n) a \cdot \nabla\nabla\theta^n \\ & + 2 |\nabla\theta^n|^{-4} (a\nabla\theta^n : \nabla\nabla\theta^n)(b \cdot \nabla\theta^n) \nabla\theta^n \end{aligned} \tag{53}$$

when b is perpendicular to $\nabla\theta^n$ (e.g. when $b = v^n$), this expression simplifies into

$$a \cdot \nabla P^n b = - |\nabla\theta^n|^{-2} (ab : \nabla\nabla\theta^n) \nabla\theta^n \quad (b \perp \nabla\theta^n) \tag{54}$$

whereas when $b = \nabla\theta^n$ it becomes

$$a \cdot \nabla P^n \nabla\theta^n = |\nabla\theta^n|^{-2} (a\nabla\theta^n : \nabla\nabla\theta^n) \nabla\theta^n - a \cdot \nabla\nabla\theta^n \tag{55}$$

we also have (using $\nabla\theta^n \cdot v^n = 0$)

$$\begin{aligned} v^n v^n : \nabla\nabla P^n v^n = {} & - |\nabla\theta^n|^{-2} (v^n v^n v^n \vdots \nabla\nabla\nabla\theta^n) \nabla\theta^n \\ & - 2 |\nabla\theta^n|^{-2} (v^n v^n : \nabla\nabla\theta^n) v^n \cdot \nabla\nabla\theta^n \\ & + 4 |\nabla\theta^n|^{-4} (v^n \nabla\theta^n : \nabla\nabla\theta^n)(v^n v^n : \nabla\nabla\theta^n) \nabla\theta^n \end{aligned} \tag{56}$$

and (these identities holds almost surely under iterations within the scheme)

$$\begin{aligned} P^n \left(\tfrac{1}{2}\xi^n + \tfrac{1}{2\sqrt{3}} \eta^n \right) \cdot \nabla P^n \xi^n = {} & \mathbb{E} \left(P^n \left(\tfrac{1}{2}\xi^n + \tfrac{1}{2\sqrt{3}} \eta^n \right) \cdot \nabla P^n \xi^n \right) \\ = {} & - \tfrac{1}{2} |\nabla\theta^n|^{-2} (P^n : \nabla\nabla\theta^n) \nabla\theta^n \end{aligned} \tag{57}$$

$$\begin{aligned} & - |\nabla\theta^n|^{-2} \left(\left(\tfrac{1}{2}\xi^n + \tfrac{1}{2\sqrt{3}} \eta^n \right) \cdot \nabla\theta^n \right) P^n \left(\tfrac{1}{2}\xi^n + \tfrac{1}{2\sqrt{3}} \eta^n \right) \cdot \nabla P^n \nabla\theta^n \\ & = P^n \left(\tfrac{1}{2}\xi^n + \tfrac{1}{2\sqrt{3}} \eta^n \right) \cdot \nabla P^n (P^n - 1) \left(\tfrac{1}{2}\xi^n + \tfrac{1}{2\sqrt{3}} \eta^n \right) \\ & = \mathbb{E} \left(P^n \left(\tfrac{1}{2}\xi^n + \tfrac{1}{2\sqrt{3}} \eta^n \right) \cdot \nabla P^n (P^n - 1) \left(\tfrac{1}{2}\xi^n + \tfrac{1}{2\sqrt{3}} \eta^n \right) \right) = 0. \end{aligned} \tag{58}$$

We now use these relations to expand the terms in the equation for v^{n+1} in (46) and show that is identical to (52) to $O(h^2)$. First notice that (this is the first term on the second line in (52))

$$h v^n \cdot \nabla P^n v^n = -h |\nabla\theta^n|^{-2} (v^n v^n : \nabla\nabla\theta^n) \nabla\theta^n, \tag{59}$$

and hence all the leading order terms of order $O(h)$ and lower appear in both (46) and (52): these are

$$v^n + h P^n f^n - h\gamma v^n + \sqrt{h}\sigma P^n \xi^n - h |\nabla\theta^n|^{-2} (v^n v^n : \nabla\nabla\theta^n) \nabla\theta^n \tag{60}$$

The term $-\gamma P^n A^n$ also appears in both (46) and (52) (first on the third line in (46) and last at the right hand-side on the first line in (52)). Next notice that (in (52) these are the third term at the right hand-side on the first line, the second and third terms on the second line, the terms on the third line, and the first term on the fifth line):

$$\begin{aligned} & \tfrac{1}{2} h^2 P^n v^n \cdot \nabla f^n + h^2 v^n \cdot \nabla P^n f^n - h^2 \gamma v^n \cdot \nabla P^n v^n \\ & \quad + h^{3/2} \sigma v^n \cdot \nabla P^n \xi^n - |\nabla\theta^n|^{-2} (\nabla\theta^n \cdot A^n) v^n \cdot \nabla P^n \nabla\theta^n \\ & \quad + P^n A^n \cdot \nabla P^n v^n \\ & \qquad = \tfrac{1}{2} h^2 v^n \cdot \nabla (P^n f^n) + \tfrac{1}{2} h^2 v^n \cdot \nabla P^n f^n \\ & \qquad\quad - h^2 \gamma v^n \cdot \nabla P^n v^n + h^{3/2} \sigma v^n \cdot \nabla P^n \xi^n \\ & \qquad\quad - |\nabla\theta^n|^{-2} (\nabla\theta^n \cdot A^n) v^n \cdot \nabla P^n \nabla\theta^n + P^n A^n \cdot \nabla P^n v^n \\ & \qquad = \tfrac{1}{2} h^2 v^n \cdot \nabla (P^n f^n) - \tfrac{1}{2} h^2 \gamma v^n \cdot \nabla P^n v^n \\ & \qquad\quad + h^{3/2} \sigma v^n \cdot \nabla P^n \left(\tfrac{1}{2}\xi^n - \tfrac{1}{2\sqrt{3}} \eta^n \right) + v^n \cdot \nabla P^n A^n \\ & \qquad\quad - |\nabla\theta^n|^{-2} (\nabla\theta^n \cdot A^n) v^n \cdot \nabla P^n \nabla\theta^n + P^n A^n \cdot \nabla P^n v^n \\ & \qquad = \tfrac{1}{2} h^2 v^n \cdot \nabla (P^n f^n) + \tfrac{1}{2} h^2 \gamma |\nabla\theta^n|^{-2} (v^n v^n : \nabla\nabla\theta^n) \nabla\theta^n \\ & \qquad\quad + h^{3/2} \sigma v^n \cdot \nabla P^n) + v^n \cdot \nabla P^n A^n \\ & \qquad\quad - |\nabla\theta^n|^{-2} (\nabla\theta^n \cdot A^n) v^n \cdot \nabla P^n \nabla\theta^n + P^n A^n \cdot \nabla P^n v^n \\ & \qquad = \tfrac{1}{2} h^2 v^n \cdot \nabla (P^n f^n) + \tfrac{1}{2} h^2 \gamma |\nabla\theta^n|^{-2} (v^n v^n : \nabla\nabla\theta^n) \nabla\theta^n \\ & \qquad\quad + h^{3/2} \sigma v^n \cdot \nabla P^n \left(\tfrac{1}{2}\xi^n - \tfrac{1}{2\sqrt{3}} \eta^n \right) \\ & \qquad\quad - 2 |\nabla\theta^n|^{-2} (v^n (P^n A^n) : \nabla\nabla\theta^n) \nabla\theta^n \end{aligned} \tag{61}$$

where in the last equality we used (using (54) and since both v^n and $P^n A^n$ are perpendicular to $\nabla\theta^n$):

$$\begin{aligned} & v^n \cdot \nabla P^n A^n - |\nabla\theta^n|^{-2} (\nabla\theta^n \cdot A^n) v^n \cdot \nabla P^n \nabla\theta^n + P^n A^n \cdot \nabla P^n v^n \\ & \quad = v^n \cdot \nabla P^n A^n + v^n \cdot \nabla P^n (P^n - 1) A^n + P^n A^n \cdot \nabla P^n v^n \\ & \quad = v^n \cdot \nabla P^n P^n A^n + P^n A^n \cdot \nabla P^n v^n \\ & \quad = -2 |\nabla\theta^n|^{-2} (v^n (P^n A^n) : \nabla\nabla\theta^n) \nabla\theta^n \end{aligned} \tag{62}$$

The terms in (61) are the second on the second line, the second on the third line, the one on fourth line and the one on the sixth line in (46).

From (57) and (58) we also have (this is the second term on the fifth line and the one on the sixth line in (52)):

$$h^2\sigma^2 P^n\left(\tfrac{1}{2}\xi^n+\tfrac{1}{2\sqrt{3}}\eta^n\right)\cdot\nabla P^n\xi^n - h^2\sigma^2|\nabla\theta^n|^{-2}$$
$$\times\left(\left(\tfrac{1}{2}\xi^n+\tfrac{1}{2\sqrt{3}}\eta^n\right)\cdot\nabla\theta^n\right)P^n\left(\tfrac{1}{2}\xi^n+\tfrac{1}{2\sqrt{3}}\eta^n\right)\cdot\nabla P^n\nabla\theta^n$$
$$=-\tfrac{1}{2}h^2\sigma^2|\nabla\theta^n|^{-2}(P^n:\nabla\nabla\theta^n)\nabla\theta^n \quad (63)$$

which is the term on the eight line in (46).

Finally, the remaining terms unaccounted for in (52) are (using (54) and (55)):

$$-\tfrac{1}{2}h^2|\nabla\theta^n|^{-2}(v^nv^n:\nabla\nabla\theta^n)v^n\cdot\nabla P^n\nabla\theta^n$$
$$=-\tfrac{1}{2}h^2|\nabla\theta^n|^{-4}(v^nv^n:\nabla\nabla\theta^n)(v^n\nabla\theta^n:\nabla\nabla\theta^n)\nabla\theta^n$$
$$+\tfrac{1}{2}h^2|\nabla\theta^n|^{-2}(v^nv^n:\nabla\nabla\theta^n)v^n\cdot\nabla\nabla\theta^n \quad (64)$$
$$-\tfrac{1}{2}h^2|\nabla\theta^n|^{-2}(v^nv^n:\nabla\nabla\theta^n)\nabla\theta^n\cdot\nabla P^nv^n$$
$$=\tfrac{1}{2}h^2|\nabla\theta^n|^{-4}(v^nv^n:\nabla\nabla\theta^n)(v^n\nabla\theta^n:\nabla\nabla\theta^n)\nabla\theta^n \quad (65)$$

and (using (56))

$$\tfrac{1}{2}h^2v^nv^n:\nabla\nabla P^nv^n$$
$$=-\tfrac{1}{2}h^2|\nabla\theta^n|^{-2}(v^nv^nv^n\vdots\nabla\nabla\nabla\theta^n)\nabla\theta^n$$
$$-h^2|\nabla\theta^n|^{-2}(v^nv^n:\nabla\nabla\theta^n)v^n\cdot\nabla\nabla\theta^n$$
$$+2h^2|\nabla\theta^n|^{-4}(v^n\nabla\theta^n:\nabla\nabla\theta^n)(v^nv^n:\nabla\nabla\theta^n)\nabla\theta^n \quad (66)$$

so that

$$-\tfrac{1}{2}h^2|\nabla\theta^n|^{-2}(v^nv^n:\nabla\nabla\theta^n)v^n\cdot\nabla P^n\nabla\theta^n$$
$$-\tfrac{1}{2}h^2|\nabla\theta^n|^{-2}(v^nv^n:\nabla\nabla\theta^n)\nabla\theta^n\cdot\nabla P^nv^n$$
$$+\tfrac{1}{2}h^2v^nv^n:\nabla\nabla P^nv^n$$
$$=-\tfrac{1}{2}h^2|\nabla\theta^n|^{-2}(v^nv^nv^n\vdots\nabla\nabla\nabla\theta^n)\nabla\theta^n$$
$$-\tfrac{1}{2}h^2|\nabla\theta^n|^{-2}(v^nv^n:\nabla\nabla\theta^n)v^n\cdot\nabla\nabla\theta^n$$
$$+2h^2|\nabla\theta^n|^{-4}(v^n\nabla\theta^n:\nabla\nabla\theta^n)(v^nv^n:\nabla\nabla\theta^n)\nabla\theta^n$$
$$=-\tfrac{1}{2}h^2v^n\cdot\nabla\left(|\nabla\theta^n|^{-2}(v^nv^n:\nabla\nabla\theta^n)\nabla\theta^n\right)$$
$$+h^2|\nabla\theta^n|^{-4}(v^n\nabla\theta^n:\nabla\nabla\theta^n)(v^nv^n:\nabla\nabla\theta^n)\nabla\theta^n \quad (67)$$

These are the terms on the fifth line and the one on the seventh line in (46), which were the last ones to account for in this equation (meaning also that we are finally done).

4. Concluding remarks

Regarding the numerical integration of Langevin equations with holonomic constraint, we found the literature quite confusing. We hope that the (courageous) reader who bore with us until the end will be gratified to see that, with heavy but straightforward calculations, a very simple, second-order accurate integrator for these equations can be derived. At the end, this integrator requires no more numerical effort than its deterministic counterpart. We also stress that the calculations above show that analyzing the accuracy of integrators is more complicated for Langevin equations than Hamiltonian systems. As a result, one may easily overestimate the order of accuracy of a scheme (as was done, e.g. by one of us in [9]).

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